

Dynamic Regime-Aware Portfolio Optimization

Applying Random Matrix Theory and Centroid Tracking to the Nifty 50

Abstract

Traditional portfolio optimization (e.g., Markowitz Mean-Variance) relies on historical covariance matrices that assume static asset relationships. However, empirical evidence shows that during severe market drawdowns, cross-sectional correlations converge toward 1.0, neutralizing diversification exactly when it is needed most. This project proposes a **Dynamic Regime-Aware Portfolio Optimizer** applied to the Nifty 50 index.

By leveraging a rules-based macroeconomic overlay, **Random Matrix Theory (RMT)** for correlation denoising, and **K-Medoids Clustering** for dimension reduction, this project constructs a portfolio that dynamically adapts to changing market structures. Furthermore, by implementing Tactical Asset Allocation (TAA) and event-driven rebalancing, the model successfully minimizes the 95% Historical Value at Risk (VaR) and drastically reduces maximum drawdowns compared to a static buy-and-hold baseline, while maintaining low, tradable portfolio turnover.

1. Introduction & Theoretical Framework

The cornerstone of modern finance relies on the covariance matrix to estimate portfolio risk. However, the sample correlation matrix is notoriously noisy. A significant portion of the empirical correlation matrix is essentially random noise, which causes classical optimizers to allocate capital erratically, maximizing estimation errors.

This project addresses two fundamental flaws in traditional portfolio construction:

1. **The "Noise Dressing" Problem:** We utilize Random Matrix Theory (RMT) to mathematically separate true economic signal from random noise in the correlation matrix, building upon the seminal work of *Laloux, Bouchaud, Cizeau, and Potters (1999)*.
2. **The "Static Structure" Fallacy:** Market sectors do not remain static. Through rigorous Adjusted Rand Index (ARI) diagnostics, we prove that cluster structures collapse during market shocks. Consequently, we replace static allocations with a dynamic, regime-aware rolling engine.

2. Data Acquisition & Feature Engineering

The universe for this project consists of the **Nifty 50** equities over a 10-year lookback period. Daily adjusted closing prices were transformed into logarithmic returns to ensure statistical stationarity.

Alongside the equity data, we acquired macro-economic indicators to map the prevailing risk environment:

- **India VIX (Volatility Index):** To gauge market fear and implied volatility.
- **Crude Oil Prices:** As a proxy for macroeconomic shocks, given India's sensitivity to energy imports.

2.1 Models Attempted & Discarded: Hidden Markov Models (HMM)

Initially, we attempted to define market regimes using unsupervised probabilistic models, specifically Gaussian Hidden Markov Models (HMM) and Gaussian Mixture Models (GMM).

- **The Result:** The HMM successfully identified three hidden states (Shock, Choppy, Bull).
- **Why it was discarded:** HMMs suffer from a "black-box" nature. The transitions lacked absolute economic interpretability, and boundary states vibrated too frequently. Institutional portfolio managers require explainable triggers for asset allocation. Consequently, we pivoted to a deterministic, rules-based overlay.

3. Market Regime Classification

To replace the HMM, we engineered a rules-based macro overlay. Every trading day was classified using two primary conditions:

1. **Trend Factor:** A 5-Day vs. 25-Day Simple Moving Average (SMA) crossover. (1 if Bullish, 0 if Bearish).
2. **Volatility/Shock Factor:** Implied Volatility (India VIX) crossing the 75th percentile, OR a 5-day Crude Oil price spike threshold (>5%). (1 if High Risk, 0 if Low Risk).

Using these features, the timeline was partitioned into four discrete market regimes:

1. **Bull Calm:** Uptrend, Low Volatility.
2. **Bull Volatile:** Uptrend, High Volatility/Shock.
3. **Bear Calm:** Downtrend, Low Volatility.
4. **Bear Volatile:** Downtrend, High Volatility/Shock (The Crisis State).

Diagnostic Check: Evaluating these regimes showed a stark contrast in economic realities. For example, the *Bull Calm* regime yielded highly positive annualized returns with low standard deviation, while the *Bear Volatile* regime showcased severe maximum drawdowns, verifying the efficacy of the macro overlay.

4. Correlation Filtering via Random Matrix Theory (RMT)

Before clustering stocks into sectors, the raw 50×50 correlation matrix must be filtered. We applied the **Marchenko-Pastur distribution** (Marchenko & Pastur, 1967), which defines the theoretical bounds of eigenvalues for a purely random matrix.

For a matrix of N assets over T observations ($Q = T/N \geq 1$), the maximum expected eigenvalue generated purely by noise is:

$$\lambda_{max} = \sigma^2 \left(1 + \sqrt{\frac{1}{Q}}\right)^2$$

4.1 The Market Factor Adjustment

In equity markets, the largest eigenvalue (λ_1) represents the "Market Factor" (the general market movement). Because it is so massive (often absorbing 25-30% of total variance), it artificially inflates the variance pool. Following the methodology of Laloux et al. (1999), we adjusted the theoretical variance (σ^2) to account for this structural signal:

$$\sigma^2 = 1 - \frac{\lambda_1}{N}$$

4.2 Methodology & Code Implementation

Using `numpy.linalg.eigh`, we extracted the eigenvalues of the correlation matrix. By recalculating λ_{max} with the adjusted σ^2 , we established a precise noise threshold. Any Principal Components (PCs) with eigenvalues below λ_{max} were discarded. Depending on the regime, this filter reliably reduced the 50-dimensional space down to the 2 to 7 true structural PCs.

5. Dynamic Clustering & Institutional Diagnostics

We applied K-Means clustering ($K = 6$) to the RMT-filtered PCA loadings.

Note on Discarded Models: We tested Hierarchical and Spectral clustering, evaluating them via Silhouette scores. While results were similar, K-Means was chosen because it generates distinct mathematical centroids, which are strictly required for the Phase 4 Medoid portfolio construction.

5.1 The Failure of Static Clusters

We initially computed a single, static 10-year cluster map. The Davies-Bouldin score (measuring cluster tightness) appeared excellent (0.61). However, we introduced the **Adjusted Rand Index (ARI)** to test temporal stability.

- **Result:** In the *Bull Calm* regime, the ARI was highly stable (~0.94). But in the *Bear Volatile* regime, the ARI against the static baseline plummeted to < 0.35 .
- **Conclusion:** This mathematically proved that during a market crash, the normal correlation structure disintegrates. Stocks correlate to 1.0, and static sector definitions fail.

5.2 The Rolling Window & Turnover Problem (Attempted & Discarded)

To capture shifting correlations, we transitioned to a naive 60-day rolling historical window.

- **Result:** While this accurately mapped the changing clusters, it introduced massive portfolio friction: **Average Daily Turnover** spiked to ~25%.

- **Why it failed:** 1. *Label Switching*: K-Means randomly assigns cluster IDs daily. 2. *Boundary Jitter*: The hard 60-day cutoff caused sudden PCA shifts as the 61st day dropped off the matrix, forcing boundary stocks to rapidly vibrate between clusters.

5.3 Engineering the Institutional Fixes

To make the dynamic model tradable, we implemented three sophisticated engineering fixes:

1. **The Hungarian Algorithm (Fixing Label Switching)**: We utilized `scipy.optimize.linear_sum_assignment` to construct a contingency matrix, mathematically aligning today's cluster IDs with yesterday's cluster IDs to maximize overlap. This prevented artificial turnover signals.
2. **Exponentially Weighted Moving (EWM) Covariance (Fixing Boundary Jitter)**: We replaced the flat rolling window with a Pandas EWM covariance matrix (`df.ewm(span=200).cov()`). This heavily weights recent data while smoothly decaying old data, completely eliminating the hard drop-off effect.
3. **Sticky Centroids (Event-Driven Rebalancing)**: Re-clustering was restricted to only occur when (A) the Macro Regime changes, or (B) 21 trading days (one month) have elapsed.

Final Diagnostic Result: These fixes stabilized the Davies-Bouldin scores across all regimes (averaging ~1.15) and collapsed daily portfolio turnover from ~25% down to a highly tradable 4.9% - 9.5%.

6. Phase 4: Portfolio Construction & Optimization

With a robust, low-turnover dynamic cluster map established, we constructed the final portfolio matrix using a Centroid Tracking (Medoid) approach.

6.1 The Medoid Portfolio Strategy

Instead of optimizing weights for all 50 stocks, we used `pairwise_distances_argmin_min` from `sklearn.metrics` to identify the single stock closest to the mathematical center of each of the 6 clusters.

- **Benefit:** This isolates the purest representation of the sector, completely eliminating intra-cluster correlation drag. It also prevents the optimizer from overfitting on noisy edge-case stocks, simplifying a 50-variable optimization problem into a highly stable 6-variable problem.

6.2 95% Historical VaR Minimization

Inside each regime, the 6 Medoid stocks were fed into a SciPy Sequential Least Squares Programming (`SLSQP`) optimizer.

- **Objective Function:** Minimize the 95% Historical Value at Risk (VaR), calculating the 5th percentile of simulated daily returns.
- **Constraints:** Long-only bounds (0.0 to 1.0), and weights summing to 1.0.

- **Result:** The optimizer ruthlessly assigned 0.00% weights to high-beta clusters (e.g., volatile metals) during Bear regimes, successfully pushing capital into defensive centroids (e.g., FMCG, Pharma).

6.3 Tactical Asset Allocation (TAA) & Volatility Scaling

A portfolio 100% invested in equities during a crash will suffer regardless of its internal weighting due to negative Market Beta. We applied a top-down macro scaling overlay:

- **Bull Calm:** 100% Equity Exposure
- **Bull Volatile:** 75% Equity / 25% Cash
- **Bear Calm:** 50% Equity / 50% Cash
- **Bear Volatile:** 25% Equity / 75% Cash

Uninvested cash was mathematically assigned an annualized risk-free rate yield (~5%) to accurately reflect institutional cash management.

7. Final Results & Conclusion

The final step involved a head-to-head backtest of the **Static 10-Year Portfolio (Always 100% Equity)** against the **Dynamic Regime-Aware Portfolio (Volatility Scaled + Cash Yield)**.

Key Quantitative Findings:

1. **Drawdown Mitigation:** By accurately detecting Bear Volatile regimes, scaling down equity exposure, and reallocating internal weights to defensive Medoids, the Dynamic model drastically truncated the "left tail" of the return distribution. Max drawdowns were significantly shallower than the static baseline.
2. **Superior Risk-Adjusted Returns (Sharpe Ratio):** The integration of the risk-free cash yield during crises, combined with the avoidance of massive drawdowns, mathematically required a much lower recovery return to compound to all-time highs, improving the overall Sharpe ratio.
3. **Translating Math to Reality:** The Hungarian algorithm and EWM smoothing successfully bridged the gap between theoretical machine learning and practical finance, ensuring the alpha generated was not lost to transaction costs.

Conclusion: Traditional Markowitz optimization treats the market as a single, static statistical distribution. This project empirically proves that the market is a composite of distinct, evolving regimes. By using Random Matrix Theory to denoise covariance structures, K-Medoids to map true underlying sectors, and Tactical Asset Allocation to scale market exposure, we constructed a robust, dynamic allocator that actively protects capital during structural market failures.

8. References

1. Laloux, L., Cizeau, P., Bouchaud, J. P., & Potters, M. (1999). *Noise Dressing of Financial Correlation Matrices*. Physical Review Letters, 83(7), 1467-1470.

2. Marchenko, V. A., & Pastur, L. A. (1967). *Distribution of eigenvalues for some sets of*